Take-Home Case Study: Data & AI Engineer (Growth & Analytics)

At a glance

You receive	This brief and a data folder with three CSV files
Time allowed	48 hours from when you receive this brief. Expected effort is about 3 hours
Your tasks	Part 1: Analyse · Part 2: Recommend · Part 3: Automate and make it self-serve · Final step: Short reflection
Submit	One link, as explained in the email you received with this brief
Tools	Any tools you like. AI tools are allowed

The situation

UrbanCart is a growing D2C e-commerce brand. Between January and June 2026 it spent about ₹16 lakh acquiring customers through four paid channels: Google Ads, Meta Ads, Affiliate and Influencer. Some customers also arrive organically.

The CMO has been asked to **cut paid marketing spend by 20%** next quarter without hurting growth. She wants to know which channel to cut and why. She also asks a lot of follow-up questions every week, and wants a way to get answers herself without waiting on the data team.

You are the new Data & AI Engineer on the Growth & Marketing team. Your job has three connected parts:

1. **Analyse** the data to understand how each channel really performs.
2. **Recommend** what to cut, based on your analysis.
3. **Automate** the analysis and design a self-serve or conversational tool, so the CMO can get these answers on her own every week.

Each part builds on the one before it, so do them in order.

Your data

The data folder contains three CSV files, exported directly from UrbanCart's source systems. The data has not been cleaned.

File	Columns	Description
users.csv	user_id, signup_date, city, acquisition_channel, email	One row per user who signed up between Jan and Jun 2026
orders.csv	order_id, user_id, order_date, order_value, status	One row per order; status is delivered, cancelled or returned
marketing_spend.csv	month, channel, spend_inr	Monthly spend per paid channel

Definitions to use

- **New customer:** a user who placed at least one successful order, counted under their acquisition channel. A successful order is one with status delivered; cancelled and returned orders don't count.
- **CAC (customer acquisition cost):** total channel spend ÷ new customers from that channel.
- **Revenue per customer:** total revenue from successful orders ÷ number of customers, for the full Jan–Jun period.
- **Repeat purchase rate:** % of customers with two or more successful orders.
- **LTV (lifetime value):** revenue per customer × contribution margin. Assume a contribution margin of **30%** for all channels.
- **LTV:CAC:** LTV ÷ CAC.

You may use any tool: SQL (DuckDB, SQLite, PostgreSQL, BigQuery or other), Python, or both. If you haven't used SQL on CSV files before, DuckDB is free and needs no setup.

Part 1: Analyse

Answer these questions using SQL or Python. Show your numbers in a simple table, and keep the code you used. Round money to the nearest rupee and percentages to one decimal place.

1. **Data quality.** Before analysing, check the data. List every issue you found, how many rows it affected, and how you handled it.
2. **Acquisition cost.** For each paid channel, what were the total spend, number of new customers and CAC?
3. **Customer value.** For each channel, including Organic, what were the revenue per customer and the repeat purchase rate?
4. **Return on spend.** What is the LTV:CAC for each paid channel? Which channel gives the best and worst return?
5. **Trend.** Calculate CAC by month for each paid channel, grouping new customers by their signup month. Which channel is getting more expensive, and by how much from January to June?
6. **Sanity check.** Pick one number from your analysis and explain how you verified it is correct.

Part 2: Recommend

Using your Part 1 results, write a short memo to the CMO (**maximum 250 words**) that answers:

- Which channel or channels should UrbanCart cut, and by how much, to save 20% of paid spend?
- Which channel should be protected or grown, and why?
- What is the biggest risk of your recommendation, and what would you watch for after the cut?

Include **2 or 3 charts** that support your recommendation. Use any tool: Excel, Google Sheets, Python, Looker Studio, Power BI or similar.

Write for a busy senior leader: lead with the decision, keep it plain, and back every claim with a number.

Part 3: Automate and make it self-serve

The CMO liked your analysis and wants it every week without asking. In production, this data would come from **GA4** (website activity), **Salesforce** (customers and orders), **MoEngage** (campaign engagement) and the **ad platforms** (spend).

Write a design note (**maximum 400 words**) with a simple architecture diagram. A hand-drawn photo is fine. No cloud account is needed.

A. Automate the pipeline. How would you bring this data into one place automatically every day, on GCP *or* AWS? Cover:

- where the raw data lands, and which warehouse you would use
- how the daily job is scheduled and run
- how the data issues you found in Part 1 are caught and handled automatically
- how API keys are stored securely, and how you would know if the pipeline fails

B. Make it self-serve. Design a tool the CMO can use on her own, either a dashboard, a conversational agent where she types questions like *"What was Meta's CAC last month?"*, or both. Explain:

- what the tool shows or answers, and which tables it reads from
- how it works, step by step, from her question to the answer
- if you use an AI model: how you would make sure it gives correct numbers and doesn't make them up
- how you would test it before giving it to her

Final step: Short reflection

After finishing Parts 1–3, answer the question below in about 200–250 words.

There are no right answers. We just want to hear about you, in your own words.

This is a demanding role that requires a lot of rigor and hard work. Why do you still want to apply for it?

What to submit

Share **one link** (a GitHub repository or a Google Drive folder with view access) containing:

1. **Part 1:** your answers in a table (PDF, Excel or notebook) and all code you used
2. **Part 2:** your memo and charts
3. **Part 3:** your design note and diagram
4. **AI-use note (3–5 lines):** which AI tools you used, for what, and one thing the AI got wrong that you had to fix
5. **Reflection:** your answer to the final-step question

How to submit

1. Put everything in one folder, with files named clearly by part, for example *Part1_Analysis*, *Part2_Memo*, *Part3_Design*.
2. Check that the link opens for anyone with it, without needing to request access.
3. Send the link as explained in the email you received with this brief, before the deadline.

Rules

- **Deadline:** 48 hours from when you receive this brief. If you need more time for a genuine reason, let us know before the deadline.
- **Expected effort:** about 3 hours. You don't need to use the full 48 hours.

- **Tools:** use any tools you like. AI tools such as Claude, ChatGPT or Copilot are allowed and encouraged, but you must be able to explain every part of your work.
- **Questions:** reply to the email you received this brief with if anything is unclear. If you make an assumption, write it down.
- **Own work:** this is an individual assignment. Shortlisted candidates will be asked to walk through their submission live.

How you'll be evaluated

We look for:

- **Rigor:** correct numbers, careful data checks and clearly stated assumptions
- **Thinking:** a clear, well-reasoned recommendation backed by data
- **Engineering sense:** a practical, reliable design for automation and the self-serve tool
- **Smart use of AI:** using AI to move faster while checking its output
- **Ownership and effort:** a complete, tidy, on-time submission, with care in the details

We value a correct, well-explained answer over a long one. Good luck!